Machine-Level Programming II: Control

15-213: Introduction to Computer Systems
6th Lecture, Sept. 14, 2017

Today's Instructor:
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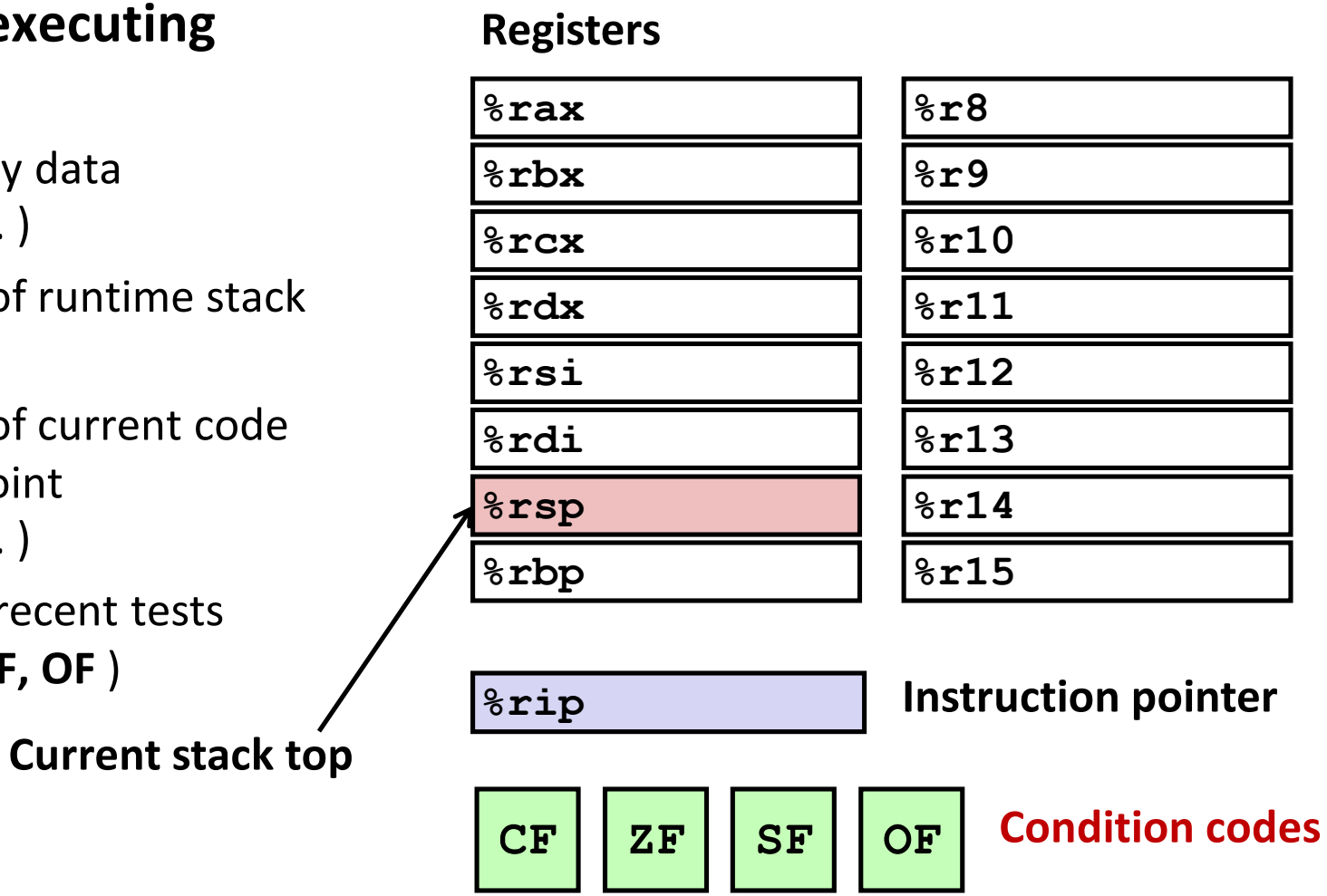
Today

- Control: Condition codes
- Conditional branches
- Loops
- Switch Statements

Processor State (x86-64, Partial)

- **Information about currently executing program**
  - Temporary data (**%rax**, …)
  - Location of runtime stack (**%rsp**)
  - Location of current code control point (**%rip**, …)
  - Status of recent tests (**CF, ZF, SF, OF**)



Condition Codes (Implicit Setting)

- **Single bit registers**
  - **CF** Carry Flag (for unsigned) **SF** Sign Flag (for signed)
  - **ZF** Zero Flag **OF** Overflow Flag (for signed)

- **Implicitly set (as side effect) of arithmetic operations**

  Example: **`addq`** *Src,Dest* ↔ **`t = a+b`**

  CF set if carry/borrow out from most significant bit (unsigned overflow)

  ZF set if **`t == 0`**

  SF set if **`t < 0`** (as signed)

  OF set if two's-complement (signed) overflow

  `(a>0 && b>0 && t<0) || (a<0 && b<0 && t>=0)`

- **Not set by `leaq` instruction**

CF set when

```
  1xxxxxxxxxxxx...
+ 1xxxxxxxxxxxx...
-----------------
1 xxxxxxxxxxxxx...
```

Carry

```
1 0xxxxxxxxxxxx...
- 1xxxxxxxxxxxx...
-----------------
  1xxxxxxxxxxxx...
```

Borrow

For unsigned arithmetic, this reports overflow

SF set when

```
  yxxxxxxxxxxxx...
+ yxxxxxxxxxxxx...
------------------
  1xxxxxxxxxxxx...
```

For signed arithmetic, this reports when result is a negative number

OF set when

```
  yxxxxxxxxxxxx...
+ yxxxxxxxxxxxx...
------------------
  zxxxxxxxxxxxx...
```

z = ~y

For signed arithmetic, this reports overflow

ZF set when

000000000000…00000000000

定点算术运算指令汇总（32位）

指令	显式操作数	影响的常用标志	操作数类型	AT&T 指令助记符	对应 C 运算符
ADD	2 个	OF、ZF、SF、CF	无/带符号整数	addb、addw、addl	+
SUB	2 个	OF、ZF、SF、CF	无/带符号整数	subb、subw、subl	−
INC	1 个	OF、ZF、SF	无/带符号整数	incb、incw、incl	++
DEC	1 个	OF、ZF、SF	无/带符号整数	decb、decw、decl	−−
NEG	1 个	OF、ZF、SF、CF	无/带符号整数	negb、negw、negl	−
CMP	2 个	OF、ZF、SF、CF	无/带符号整数	cmpb、cmpw、cmpl	<, <=, >, >=
MUL	1 个	无	无符号整数	mulb、mulw、mull	*
MUL	2 个	无	无符号整数	mulb、mulw、mull	*
MUL	3 个	无	无符号整数	mulb、mulw、mull	*
IMUL	1 个	无	带符号整数	imulb、imulw、imull	*
IMUL	2 个	无	带符号整数	imulb、imulw、imull	*
IMUL	3 个	无	带符号整数	imulb、imulw、imull	*
DIV	1 个	无	无符号整数	divb、divw、divl	/，%
IDIV	1 个	无	带符号整数	idivb、idivw、idivl	/，%

定点加法指令举例

- 假设R[ax]=FFFAH，R[bx]=FFF0H，则执行Intel格式指令“add ax, bx”后，AX、BX中的内容各是什么？标志CF、OF、ZF、SF各是什么？要求分别将操作数作为无符号数和带符号整数解释并验证指令执行结果。

解：功能：R[ax]←R[ax]+R[bx]，指令执行后的结果如下

R[ax]=FFFAH+FFF0H=FFEAH ，BX中内容不变

CF=1，OF=0，ZF=0，SF=1

若是无符号整数运算，则CF=1说明结果溢出

验证：FFFA的真值为65535-5=65530，FFF0的真值为65515

FFEA的真值为65535-21=65514≠65530+65515，即溢出

若是带符号整数运算，则OF=0说明结果没有溢出

验证：FFFA的真值为-6，FFF0的真值为-16

FFEA的真值为-22=-6+(-16)，结果正确，无溢出

定点乘法指令举例

- 假设R[eax]=000000B4H，R[ebx]=00000011H，M[000000F8H]=000000A0H，请问：

  (1) 执行指令“mulb %bl”后，哪些寄存器的内容会发生变化？是否与执行“imulb %bl”指令所发生的变化一样？为什么？请用该例给出的数据验证你的结论。

解：“mulb %bl”功能为 R[ax]←R[al]×R[bl]，执行结果如下

R[ax]=B4H*11H（无符号整数180和17相乘）

R[ax]=0BF4H，真值为3060=180 × 17

```
          1011 0100
     x    0001 0001
     ---------------
          1011 0100
     1011 0100
     ---------------
0000 1011 1111 0100
```

AH= ?　AL= ?

“imulb %bl”功能为 R[ax]←R[al]×R[bl]

R[ax]=B4H*11H（带符号整数-76和17相乘）

若R[ax]=0BF4H，则真值为3060≠-76 × 17

R[al]=F4H, R[ah]=? AH中的变化不一样！

R[ax]=FAF4H, 真值为-1292=-76 × 17

对于带符号乘，若积只取低n位，则和无符号相同；若取2n位，则采用“布斯”乘法

定点乘法指令举例

- 假设R[eax]=000000B4H，R[ebx]=00000011H，M[000000F8H]=000000A0H，请问：

(2) 执行指令“imull $-16, (%eax,%ebx,4), %eax”后哪些寄存器和存储单元发生了变化？乘积的机器数和真值各是多少？

解：“imull -16, (%eax,%ebx,4),%eax”

功能为 R[eax]←(-16)×M[R[eax]+R[ebx]×4] ，执行结果如下

R[eax]+R[ebx]×4=000000B4H+00000011H<<2=000000F8H

R[eax]=(-16)×M[000000F8H]

=(-16)× 000000A0H（带符号整数乘）

=FFFFFF60H<<4

=FFFFF600H

EAX中的真值为-2560

SKIP

整数乘除指令

- 乘法指令：可给出一个、两个或三个操作数
  - 若给出一个操作数SRC，则另一个源操作数隐含在AL/AX/EAX中，将SRC和累加器内容相乘，结果存放在AX（16位）或DX-AX（32位）或EDX-EAX（64位）中。DX-AX表示32位乘积的高、低16位分别在DX和AX中。
  - 若指令中给出两个操作数DST和SRC，则将DST和SRC相乘，结果在DST中。
  - 若指令中给出三个操作数REG、SRC和IMM，则将SRC和立即数IMM相乘，结果在REG中。
- 除法指令：只明显指出除数，用EDX-EAX中内容除以指定的除数
  - 若为8位，则16位被除数在AX寄存器中，商送回AL，余数在AH
  - 若为16位，则32位被除数在DX-AX寄存器中，商送回AX，余数在DX
  - 若为32位，则被除数在EDX-EAX寄存器中，商送EAX，余数在EDX

以上内容不要死记硬背，遇到具体指令时能查阅到并理解即可。 BACK

按位运算指令举例

假设short型变量x被编译器分配在寄存器AX中，R[ax]=FF80H，则以下汇编代码段执行后变量x的机器数和真值分别是多少？

```
movw %ax, %dx
salw   $2, %ax       1111 1111 1000 0000<<2
addl   %dx, %ax      1111 1111 1000 0000+1111 1110 0000 0000
sarw   $1, %ax       1111 1101 1000 0000>>1=1111 1110 1100 0000
```

解：$2和$1分别表示立即数2和1 。

x是short型变量，故都是算术移位指令，并进行带符号整数加。假设上述代码段执行前R[ax]=x，则执行((x<<2)+x)>>1后，R[ax]=5x/2。算术左移时，AX中的内容在移位前、后符号未发生变化，故OF=0，没有溢出。最终AX的内容为FEC0H，解释为short型整数时，其值为-320。验证：x=-128，5x/2=-320。经验证，结果正确。

Condition Codes (Explicit Setting: Compare)

- **Explicit Setting by Compare Instruction**
  - `cmpq` *Src2, Src1*
  - `cmpq b,a` like computing `a-b` without setting destination

  - **CF set** if carry/borrow out from most significant bit (used for unsigned comparisons)
  - **ZF set** if `a == b`
  - **SF set** if `(a-b) < 0` (as signed)
  - **OF set** if two's-complement (signed) overflow
    `(a>0 && b<0 && (a-b)<0) || (a<0 && b>0 && (a-b)>0)`

Condition Codes (Explicit Setting: Test)

- **Explicit Setting by Test instruction**
  - `testq` *Src2, Src1*
    - `testq b,a` like computing `a&b` without setting destination
  - Sets condition codes based on value of *Src1* & *Src2*
  - Useful to have one of the operands be a mask
  - **ZF set** when `a&b == 0`
  - **SF set** when `a&b < 0`

Very often:

```
testq  %rax,%rax
```

Reading Condition Codes

- **SetX Instructions**
  - Set low-order byte of destination to 0 or 1 based on combinations of condition codes
  - Does not alter remaining 7 bytes

SetX	Condition	Description
`sete`	`ZF`	Equal / Zero
`setne`	`~ZF`	Not Equal / Not Zero
`sets`	`SF`	Negative
`setns`	`~SF`	Nonnegative
`setg`	`~(SF^OF)&~ZF`	Greater (Signed)
`setge`	`~(SF^OF)`	Greater or Equal (Signed)
`setl`	`(SF^OF)`	Less (Signed)
`setle`	`(SF^OF)|ZF`	Less or Equal (Signed)
`seta`	`~CF&~ZF`	Above (unsigned)
`setb`	`CF`	Below (unsigned)

x86-64 Integer Registers

%rax	%al	%r8	%r8b
%rbx	%bl	%r9	%r9b
%rcx	%cl	%r10	%r10b
%rdx	%dl	%r11	%r11b
%rsi	%sil	%r12	%r12b
%rdi	%dil	%r13	%r13b
%rsp	%spl	%r14	%r14b
%rbp	%bpl	%r15	%r15b

- Can reference low-order byte

Reading Condition Codes (Cont.)

- **SetX Instructions:**
  - Set single byte based on combination of condition codes
- **One of addressable byte registers**
  - Does not alter remaining bytes
  - Typically use **`movzbl`** to finish job
    - 32-bit instructions also set upper 32 bits to 0

```
int gt (long x, long y)
{
  return x > y;
}
```

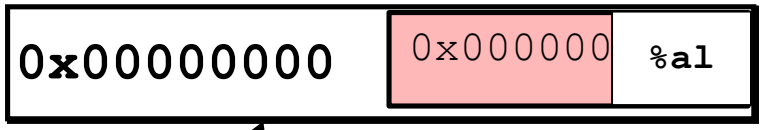
Register	Use(s)
`%rdi`	Argument **`x`**
`%rsi`	Argument **`y`**
`%rax`	Return value

```
cmpq   %rsi, %rdi    # Compare x:y
setg   %al           # Set when >
movzbl %al, %eax     # Zero rest of %rax
ret
```

Reading Condition Codes (Cont.)

Beware weirdness **movzbl** (and others)

```
movzbl %al, %eax
```



Zapped to all 0's

Use(s)
Argument **x**
Argument **y**
Return value

```
cmpq    %rsi, %rdi    # Compare x:y
setg    %al           # Set when >
movzbl  %al, %eax     # Zero rest of %rax
ret
```

Today

- Control: Condition codes
- **Conditional branches**
- Loops
- Switch Statements

Jumping

- **jX Instructions**
  - Jump to different part of code depending on condition codes

jX	Condition	Description
`jmp`	`1`	Unconditional
`je`	`ZF`	Equal / Zero
`jne`	`~ZF`	Not Equal / Not Zero
`js`	`SF`	Negative
`jns`	`~SF`	Nonnegative
`jg`	`~(SF^OF)&~ZF`	Greater (Signed)
`jge`	`~(SF^OF)`	Greater or Equal (Signed)
`jl`	`(SF^OF)`	Less (Signed)
`jle`	`(SF^OF)|ZF`	Less or Equal (Signed)
`ja`	`~CF&~ZF`	Above (unsigned)
`jb`	`CF`	Below (unsigned)

Conditional Branch Example (Old Style)

- **Generation**

```
shark> gcc -Og -S -fno-if-conversion control.c
```

Get to this shortly

```
long absdiff
  (long x, long y)
{
  long result;
  if (x > y)
    result = x-y;
  else
    result = y-x;
  return result;
}
```

```
absdiff:
    cmpq    %rsi, %rdi   # x:y
    jle     .L4
    movq    %rdi, %rax
    subq    %rsi, %rax
    ret
.L4:        # x <= y
    movq    %rsi, %rax
    subq    %rdi, %rax
    ret
```

Register	Use(s)
%rdi	Argument **x**
%rsi	Argument **y**
%rax	Return value

Expressing with Goto Code

- C allows `goto` statement
- Jump to position designated by label

```
long absdiff
  (long x, long y)
{
    long result;
    if (x > y)
        result = x-y;
    else
        result = y-x;
    return result;
}
```

```
long absdiff_j
  (long x, long y)
{
    long result;
    int ntest = x <= y;
    if (ntest) goto Else;
    result = x-y;
    goto Done;
 Else:
    result = y-x;
 Done:
    return result;
}
```

General Conditional Expression Translation (Using Branches)

C Code

```
val = Test ? Then_Expr : Else_Expr;
```

```
val = x>y ? x-y : y-x;
```

Goto Version

```
  ntest = !Test;
  if (ntest) goto Else;
  val = Then_Expr;
  goto Done;
Else:
  val = Else_Expr;
Done:
  . . .
```

- Create separate code regions for then & else expressions
- Execute appropriate one

Using Conditional Moves

- **Conditional Move Instructions**
  - Instruction supports:

    if (Test) Dest ← Src
  - Supported in post-1995 x86 processors
  - GCC tries to use them
    - But, only when known to be safe
- **Why?**
  - Branches are very disruptive to instruction flow through pipelines
  - Conditional moves do not require control transfer

C Code

```
val = Test
    ? Then_Expr
    : Else_Expr;
```

Goto Version

```
result = Then_Expr;
eval = Else_Expr;
nt = !Test;
if (nt) result = eval;
return result;
```

Conditional Move Example

```
long absdiff(long x, long y)
{
    long result;
    if (x > y)
        result = x-y;
    else
        result = y-x;
    return result;
}
```

Register	Use(s)
%rdi	Argument **x**
%rsi	Argument **y**
%rax	Return value

When is this bad?

```
absdiff:
    movq    %rdi, %rax   # x
    subq    %rsi, %rax   # result = x-y
    movq    %rsi, %rdx
    subq    %rdi, %rdx   # eval = y-x
    cmpq    %rsi, %rdi   # x:y
    cmovle  %rdx, %rax   # if <=, result = eval
    ret
```

Bad Cases for Conditional Move

Expensive Computations

```
val = Test(x) ? Hard1(x) : Hard2(x);
```

Bad Performance

- Both values get computed
- Only makes sense when computations are very simple

Risky Computations

```
val = p ? *p : 0;
```

Unsafe

- Both values get computed
- May have undesirable effects

Computations with side effects

```
val = x > 0 ? x*=7 : x+=3;
```

Illegal

- Both values get computed
- Must be side-effect free

Exercise

`cmpq b,a` like computing `a-b` without setting dest

- **CF set** if carry/borrow out from most significant bit (used for unsigned comparisons)
- **ZF set** if `a == b`
- **SF set** if `(a-b) < 0` (as signed)
- **OF set** if two's-complement (signed) overflow

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setl	(SF^OF)	Less (Signed)
setle	(SF^OF)|ZF	Less or Equal (Signed)
seta	~CF&~ZF	Above (unsigned)
setb	CF	Below (unsigned)

```
xorq    %rax, %rax
subq    $1, %rax
cmpq    $2, %rax
setl    %al
movzbl  %al, %eax
```

%rax	SF	CF	OF	ZF

Note: `setl` and `movzblq` do not modify condition codes

Exercise

`cmpq b,a` like computing `a-b` without setting dest

- **CF set** if carry/borrow out from most significant bit (used for unsigned comparisons)
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setl	(SF^OF)	Less (Signed)
setle	(SF^OF)|ZF	Less or Equal (Signed)
seta	~CF&~ZF	Above (unsigned)
setb	CF	Below (unsigned)

```
xorq    %rax, %rax
subq    $1, %rax
cmpq    $2, %rax
setl    %al
movzbl  %al, %eax
```

%rax	SF	CF	OF	ZF
0x0000 0000 0000 0000	0	0	0	1
0xFFFF FFFF FFFF FFFF	1	1	0	0
0xFFFF FFFF FFFF FFFF	1	0	0	0
0xFFFF FFFF FFFF FF01	1	0	0	0
0x0000 0000 0000 0001	1	0	0	0

Note: **`setl`** and **`movzblq`** do not modify condition codes

Today

- Control: Condition codes
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- **Loops**
- Switch Statements

"Do-While" Loop Example

C Code

```
long pcount_do
  (unsigned long x) {
  long result = 0;
  do {
    result += x & 0x1;
    x >>= 1;
  } while (x);
  return result;
}
```

Goto Version

```
long pcount_goto
  (unsigned long x) {
  long result = 0;
 loop:
  result += x & 0x1;
  x >>= 1;
  if(x) goto loop;
  return result;
}
```

- Count number of 1's in argument `x` ("popcount")
- Use conditional branch to either continue looping or to exit loop

"Do-While" Loop Compilation

```
long pcount_goto
  (unsigned long x) {
  long result = 0;
 loop:
  result += x & 0x1;
  x >>= 1;
  if(x) goto loop;
  return result;
}
```

Register	Use(s)
%rdi	Argument **x**
%rax	**result**

```
   movl    $0, %eax     #   result = 0
.L2:                    # loop:
   movq    %rdi, %rdx
   andl    $1, %edx     #   t = x & 0x1
   addq    %rdx, %rax   #   result += t
   shrq    %rdi         #   x >>= 1
   jne     .L2          #   if(x) goto loop
   rep; ret
```

General "Do-While" Translation

C Code

```
do
  Body
  while (Test);
```

Goto Version

```
loop:
  Body
  if (Test)
    goto loop
```

- **Body:**

```
{
  Statement1;
  Statement2;
    …
  Statementn;
}
```

General "While" Translation #1

- "Jump-to-middle" translation
- Used with –Og

While version

```
while (Test)
  Body
```

Goto Version

```
  goto test;
loop:
  Body
test:
  if (Test)
    goto loop;
done:
```

While Loop Example #1

C Code

```
long pcount_while
  (unsigned long x) {
  long result = 0;
  while (x) {
    result += x & 0x1;
    x >>= 1;
  }
  return result;
}
```

Jump to Middle

```
long pcount_goto_jtm
  (unsigned long x) {
  long result = 0;
  goto test;
 loop:
  result += x & 0x1;
  x >>= 1;
 test:
  if(x) goto loop;
  return result;
}
```

- Compare to do-while version of function
- Initial goto starts loop at test

General "While" Translation #2

While version

```
while (Test)
  Body
```

Do-While Version

```
  if (!Test)
    goto done;
  do
    Body
    while(Test);
done:
```

Goto Version

```
  if (!Test)
    goto done;
loop:
  Body
  if (Test)
    goto loop;
done:
```

- "Do-while" conversion
- Used with –O1

While Loop Example #2

C Code

```
long pcount_while
  (unsigned long x) {
  long result = 0;
  while (x) {
    result += x & 0x1;
    x >>= 1;
  }
  return result;
}
```

Do-While Version

```
long pcount_goto_dw
  (unsigned long x) {
  long result = 0;
  if (!x) goto done;
 loop:
  result += x & 0x1;
  x >>= 1;
  if(x) goto loop;
 done:
  return result;
}
```

- **Initial conditional guards entrance to loop**
- **Compare to do-while version of function**
  - Removes jump to middle. When is this good or bad?

"For" Loop Form

General Form

```
for (Init; Test; Update )
        Body
```

```
#define WSIZE 8*sizeof(int)
long pcount_for
  (unsigned long x)
{
  size_t i;
  long result = 0;
  for (i = 0; i < WSIZE; i++)
  {
    unsigned bit =
      (x >> i) & 0x1;
    result += bit;
  }
  return result;
}
```

Init

```
i = 0
```

Test

```
i < WSIZE
```

Update

```
i++
```

Body

```
{
  unsigned bit =
      (x >> i) & 0x1;
  result += bit;
}
```

"For" Loop → While Loop

For Version

```
for (Init; Test; Update )
        Body
```

While Version

```
Init;
while (Test) {
    Body
    Update;
}
```

For-While Conversion

Init

```
i = 0
```

Test

```
i < WSIZE
```

Update

```
i++
```

Body

```
{
  unsigned bit =
      (x >> i) & 0x1;
  result += bit;
}
```

```
long pcount_for_while
  (unsigned long x)
{
  size_t i;
  long result = 0;
  i = 0;
  while (i < WSIZE)
  {
    unsigned bit =
      (x >> i) & 0x1;
    result += bit;
    i++;
  }
  return result;
}
```

"For" Loop Do-While Conversion

C Code

```
long pcount_for
  (unsigned long x)
{
  size_t i;
  long result = 0;
  for (i = 0; i < WSIZE; i++)
  {
    unsigned bit =
      (x >> i) & 0x1;
    result += bit;
  }
  return result;
}
```

- Initial test can be optimized away

Goto Version

```
long pcount_for_goto_dw
  (unsigned long x) {
  size_t i;
  long result = 0;
  i = 0;                    Init
  if (!(i < WSIZE))
    goto done;              ! Test
 loop:
  {
    unsigned bit =
      (x >> i) & 0x1;       Body
    result += bit;
  }
  i++;  Update
  if (i < WSIZE)            Test
    goto loop;
 done:
  return result;
}
```

Today

- Control: Condition codes
- Conditional branches
- Loops
- **Switch Statements**

Switch Statement Example

```
long my_switch
   (long x, long y, long z)
{
    long w = 1;
    switch(x) {
    case 1:
        w = y*z;
        break;
    case 2:
        w = y/z;
        /* Fall Through */
    case 3:
        w += z;
        break;
    case 5:
    case 6:
        w -= z;
        break;
    default:
        w = 2;
    }
    return w;
}
```

- **Multiple case labels**
  - Here: 5 & 6
- **Fall through cases**
  - Here: 2
- **Missing cases**
  - Here: 4

Jump Table Structure

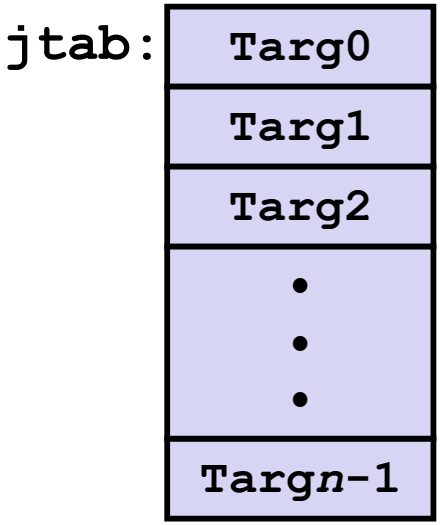
Switch Form

```
switch(x) {
  case val_0:
    Block 0
  case val_1:
    Block 1
    • • •
  case val_n-1:
    Block n–1
}
```

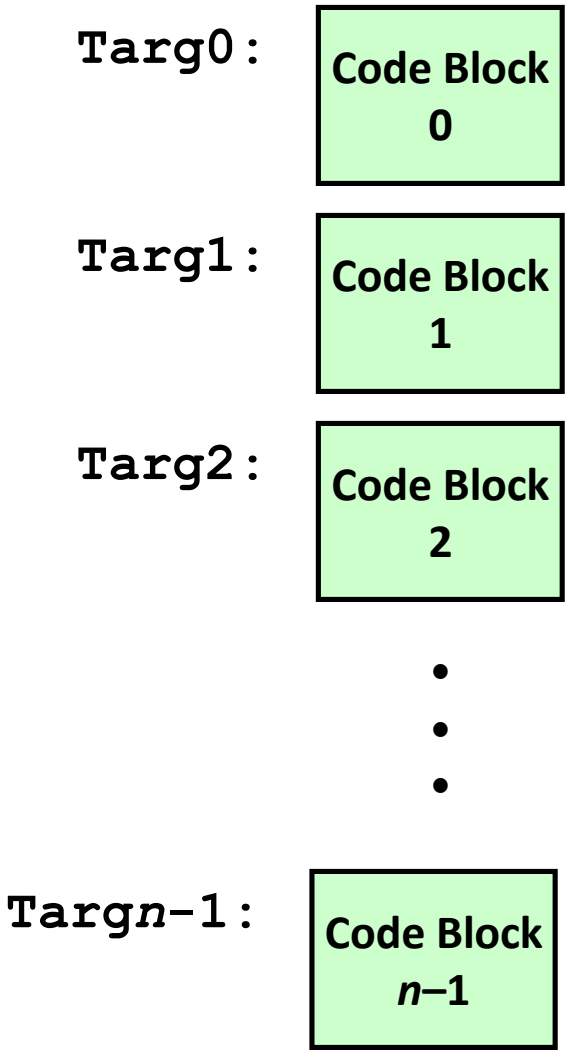
Translation (Extended C)

```
goto *JTab[x];
```

Jump Table



Jump Targets



Switch Statement Example

```
long my_switch(long x, long y, long z)
{
    long w = 1;
    switch(x) {
      . . .
    }
    return w;
}
```

Setup

```
my_switch:
    movq    %rdx, %rcx
    cmpq    $6, %rdi    # x:6
    ja      .L8
    jmp     *.L4(,%rdi,8)
```

What range of values takes default?

Register	Use(s)
%rdi	Argument **x**
%rsi	Argument **y**
%rdx	Argument **z**
%rax	Return value

Note that **w** not initialized here

Switch Statement Example

```
long my_switch(long x, long y, long z)
{
    long w = 1;
    switch(x) {
      . . .
    }
    return w;
}
```

Jump table

```
.section    .rodata
  .align 8
.L4:
  .quad     .L8   # x = 0
  .quad     .L3   # x = 1
  .quad     .L5   # x = 2
  .quad     .L9   # x = 3
  .quad     .L8   # x = 4
  .quad     .L7   # x = 5
  .quad     .L7   # x = 6
```

Setup

```
my_switch:
    movq    %rdx, %rcx
    cmpq    $6, %rdi    # x:6
    ja      .L8         # use default
    jmp     *.L4(,%rdi,8)  # goto *Jtab[x]
```

Indirect jump

Assembly Setup Explanation

- **Table Structure**
  - Each target requires 8 bytes
  - Base address at `.L4`

Jump table

```
.section    .rodata
  .align 8
.L4:
  .quad     .L8   # x = 0
  .quad     .L3   # x = 1
  .quad     .L5   # x = 2
  .quad     .L9   # x = 3
  .quad     .L8   # x = 4
  .quad     .L7   # x = 5
  .quad     .L7   # x = 6
```

- **Jumping**
  - **Direct:** `jmp .L8`
  - Jump target is denoted by label `.L8`
  - **Indirect:** `jmp *.L4(,%rdi,8)`
  - Start of jump table: `.L4`
  - Must scale by factor of 8 (addresses are 8 bytes)
  - Fetch target from effective Address `.L4 + x*8`
    - Only for $0 \le \mathbf{x} \le 6$

Jump Table

Jump table

```
.section    .rodata
  .align 8
.L4:
  .quad     .L8   # x = 0
  .quad     .L3   # x = 1
  .quad     .L5   # x = 2
  .quad     .L9   # x = 3
  .quad     .L8   # x = 4
  .quad     .L7   # x = 5
  .quad     .L7   # x = 6
```

```
switch(x) {
case 1:       // .L3
    w = y*z;
    break;
case 2:       // .L5
    w = y/z;
    /* Fall Through */
case 3:       // .L9
    w += z;
    break;
case 5:
case 6:       // .L7
    w -= z;
    break;
default:      // .L8
    w = 2;
}
```

Code Blocks (x == 1)

```
  switch(x) {
  case 1:       // .L3
        w = y*z;
        break;
   . . .
}
```

```
.L3:
   movq    %rsi, %rax   # y
   imulq   %rdx, %rax   # y*z
   ret
```

Register	Use(s)
`%rdi`	Argument **x**
`%rsi`	Argument **y**
`%rdx`	Argument **z**
`%rax`	Return value

Handling Fall-Through

```
long w = 1;
   . . .
switch(x) {
   . . .
case 2:
    w = y/z;
    /* Fall Through */
case 3:
    w += z;
    break;
   . . .
}
```

```
case 2:
        w = y/z;
        goto merge;
```

```
case 3:
        w = 1;
merge:
        w += z;
```

Code Blocks (x == 2, x == 3)

```
long w = 1;
   . . .
switch(x) {
   . . .
case 2:
    w = y/z;
    /* Fall Through */
case 3:
    w += z;
    break;
   . . .
}
```

```
.L5:                        # Case 2
   movq    %rsi, %rax
   cqto                  # sign extend
                         # rax to rdx:rax
   idivq   %rcx          #  y/z
   jmp     .L6           #  goto merge
.L9:                     # Case 3
   movl    $1, %eax      #  w = 1
.L6:                     # merge:
   addq    %rcx, %rax    #  w += z
   ret
```

Register	Use(s)
`%rdi`	Argument **x**
`%rsi`	Argument **y**
`%rcx`	**z**
`%rax`	Return value

Code Blocks (x == 5, x == 6, default)

```
  switch(x) {
    . . .
    case 5:  // .L7
    case 6:  // .L7
        w -= z;
        break;
    default: // .L8
        w = 2;
}
```

```
.L7:                    # Case 5,6
  movl   $1, %eax       #  w = 1
  subq   %rdx, %rax     #  w -= z
  ret
.L8:                    # Default:
  movl   $2, %eax       #  2
  ret
```

Register	Use(s)
`%rdi`	Argument **`x`**
`%rsi`	Argument **`y`**
`%rdx`	Argument **`z`**
`%rax`	Return value

Summarizing

- **C Control**
  - if-then-else
  - do-while
  - while, for
  - switch
- **Assembler Control**
  - Conditional jump
  - Conditional move
  - Indirect jump (via jump tables)
  - Compiler generates code sequence to implement more complex control
- **Standard Techniques**
  - Loops converted to do-while or jump-to-middle form
  - Large switch statements use jump tables
  - Sparse switch statements may use decision trees (if-elseif-elseif-else)

Summary

- **Today**
  - Control: Condition codes
  - Conditional branches & conditional moves
  - Loops
  - Switch statements